

Maths : Modular Arithmetic

$\%$ $\Rightarrow$ modulus operator

$\hookrightarrow$ remainder

$n \% a \rightarrow$ remainder when n is divided by a

$$10 \% 4 \rightarrow 2$$

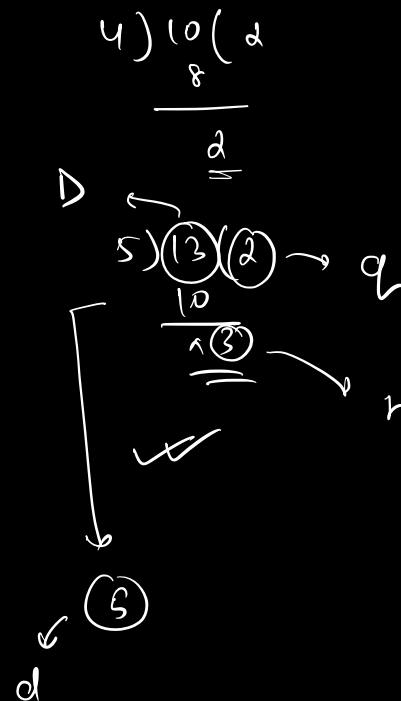
$$13 \% 5 \rightarrow 3$$

Dividend (D)

Quotient (q)

Divisor (d)

Remainder (r)



$$\text{Dividend} = \text{Quotient} \times \text{divisor} + \text{remainder}$$

$$\Rightarrow \boxed{D = qd + r}$$

$$\Rightarrow r = D - \underbrace{qd}$$

largest multiple of $d < D$

Quiz 1

$$150 \% 11 = 7$$

$$\begin{array}{r} 11 \overline{) 150} \\ \underline{11} \\ \times 40 \\ \underline{33} \\ \underline{7} \\ \hline \end{array}$$

$$r = D - qd$$

$$= 150 - \underbrace{13 \times 11}$$

largest multiple of 11 ≤ 150 .

$$= 150 - 143$$

$$= \textcircled{7}.$$

Quiz 2 $100 \% 7$

$$\begin{array}{r} \boxed{7} \overline{) 100} \\ \underline{7} \\ 30 \\ \underline{28} \\ \underline{\times 2} \\ \hline \end{array}$$

$$\text{remainder} \Rightarrow 100 - 14 \times 7$$

$$= 100 - 98$$

$$= \underline{\underline{2}}.$$

Quiz 3

largest no. $\Rightarrow -43 \quad -42 \quad -76 \quad -35$

$$-76 < -43 < -42 < -35$$

Quiz 4

$$-40 \% 7 \Rightarrow -5$$

$\rightarrow$ remainder $\Rightarrow$ Divident - largest multiple of (d) $< (D)$

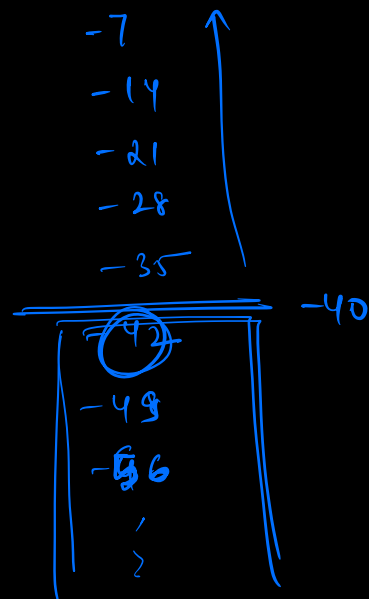
$$\text{remainder} \Rightarrow -40 - (\text{largest multiple of } 7) \\ \underline{\underline{< -40}}$$

$$-40 - (-42)$$

$$\underline{\underline{-40 > -35}}$$

$$\Rightarrow -40 \neq 42$$

$$\Rightarrow 2$$



Quiz-5

$-60 \% 9$

remänder $\Rightarrow -60 - (\text{greater multiple of } 9)$
 $9 \leq -60$

$$\Rightarrow -60 - (-63)$$

$$\Rightarrow -60 + 63$$

$$\Rightarrow \underline{3}$$

<u>9</u>	<u>-60</u>
-63	
-72	
-81	
-90	
-99	
-108	

$$a \% m \Rightarrow [0, m-1]$$

IMP

C^x | C++^x | Java^x | JS | CH | Cologan

Python ✓

$$-40 \% 7 = -5 + 7 \longrightarrow 2$$

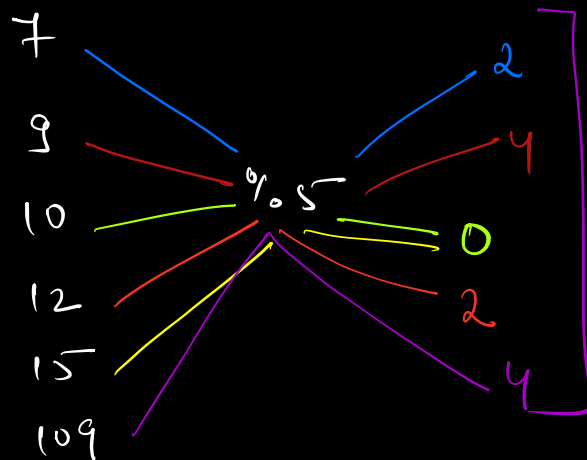
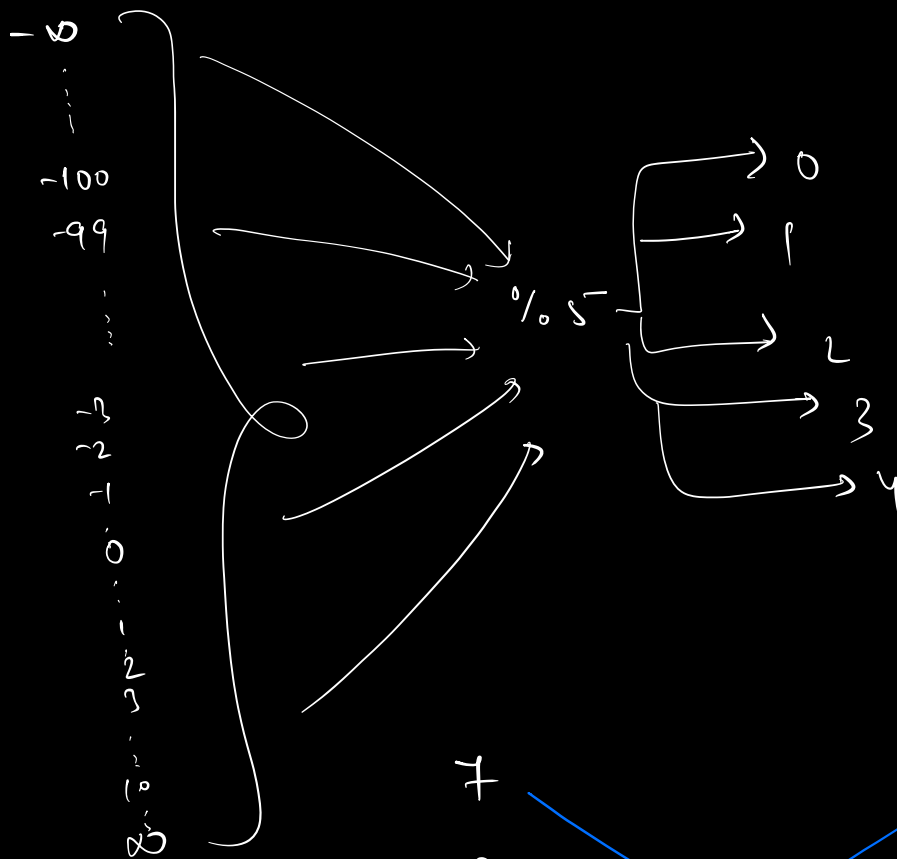
$$-60 \% 9 = -6 + 9 \longrightarrow 3$$

$$-30 \% 4 = -2 + 4 \longrightarrow 2$$

if using X | C++ | Java -

if a is -ve

$$a \% m \Rightarrow \underline{\underline{(a \% m) + m}}$$



∴ very helpful while creating :-

- Hashmap / Hashtable / Dictionary / Map
- Consistent Hashing [HLD, load balancing]
- Encryption

∴ Modular arithmetic ⇒ $(+)$ $(\times)$ $(-)$ $(\div)$
 ✓ ✓ advanced

$$\star \boxed{(a+b) \% M = (a \% M + b \% M) \% M}$$

ex ⇒ $a = 6, b = 8, M = 10$.

$(6+8) \% 10$ ⇒ $14 \% 10$ ⇒ 4	$\longleftrightarrow$ $\times$ $\longleftrightarrow$	$(6 \% 10 + 8 \% 10) \% 10$ ⇒ $(6+8) \% 10$ ⇒ $(14) \% 10$ ⇒ 4
		

Quiz-6

max^m value ⇒ $(a+b) \% M$

range ⇒ $[0, M-1] \rightarrow$ max^m $M-1$

$$\star \boxed{(a \times b) \% M = (a \% M \times b \% M) \% M}$$

$$\Rightarrow \boxed{(a \% b) \% b = a \% b}$$

★ ans ⇒ return $\underbrace{\text{ans} \% (10^9 + 7)}_{\text{why??}}$

Q1. Implement a power func, inputs a, n, p $\Rightarrow p \Rightarrow a^n \% p$.

power(a, n, p) return $a^n \% p$

ex $\Rightarrow a = 2, n = 5, p = 7$

$a^n \% p \Rightarrow 2^5 \% 7 \Rightarrow 32 \% 7 \Rightarrow 4$

$a = 3, n = 4, p = 6$

$3^4 \% 6 \Rightarrow 81 \% 6 = 3$

```
int power(a, n, p) {
```

```
    ans = 1
```

```
    for (i = 1; i <= N; i++) {
```

```
        ans = ans * a
```

```
    }
```

```
    return ans % p
```

```
}
```

$a = 10$

$n = 40$

$\xrightarrow{\quad}$

$a^n \Rightarrow 10^{40}$

+ INT x
+ LONG x
+ string
BigInt

$$(a \times b) \% M = (a \% M \times b \% M) \% M$$

$$(a \times b) \% M = (a \% M \times b \% M) \% M$$

```
int power(a, n, p) {
```

```
    ans = 1
```

```
    => for (i=1; i <= N; i++) {
```

```
        ans = (ans % p * a % p) % p
```

```
    }
```

```
    return ans
```

```
}
```

$$\begin{aligned} TC &\Rightarrow O(N) \\ SC &\Rightarrow O(1) \end{aligned}$$

ex $\Rightarrow \underbrace{2^4 \% 10}_{a=2, n=4, p=10} \Rightarrow 16 \% 10 = 6$

$$\begin{aligned} ans &= (1 \% 10 \times 2 \% 10) \% 10 \\ &= (1 \times 2) \% 10 = 2 \end{aligned}$$

$$\begin{aligned} ans &= (2 \% 10 \times 2 \% 10) \% 10 \\ &= (2 \times 2) \% 10 = 4 \end{aligned}$$

$$\begin{aligned} ans &= (4 \% 10 \times 2 \% 10) \% 10 \\ &= (4 \times 2) \% 10 = 8 \end{aligned}$$

$$\begin{aligned} ans &= (8 \% 10 \times 2 \% 10) \% 10 \\ &= 16 \% 10 = \textcircled{6} \end{aligned}$$

$$1 \leq p \leq 10^9$$

$$\text{ans} = \left(\underbrace{\text{ans} \% p}_{\substack{\text{max } p-1 \\ \checkmark}} \times \underbrace{a \% p}_{\text{max } p-1} \right) \% p$$

$$((p-1) \times (p-1)) \% p$$

$$= (10^9 \times 10^9) \% p$$

$$= \textcircled{1018} \text{ int, can still } \underline{\underline{\text{overflow}}}$$

∴ Divisibility rule

Rule of 3 : Sum of all digits is divisible by 3,
then no. also divisible by 3

$$22683 \rightarrow 2+2+6+8+3 \Rightarrow \frac{21}{3} = 0.$$

$$\begin{array}{r} 3 \overline{) 22683} \quad (7561 \\ \underline{21} \\ 16 \\ \underline{15} \\ 18 \\ \underline{18} \\ 03 \\ \underline{03} \\ 0 \end{array}$$

$$(4372) \% 3 \Rightarrow (4+3+7+2) \% 3$$

↳

$$(4 \times 10^3 + 3 \times 10^2 + 7 \times 10 + 2 \times 10^0) \% 3 \left[\begin{array}{l} (a+b) \% M \\ = (a \% M + b \% M) \% M \end{array} \right]$$

$$\rightarrow (a \times b) \% M = [a \% M \times b \% M] \% M$$

$$\Rightarrow \left[(4 \times 10^3) \% 3 + (3 \times 10^2) \% 3 + (7 \times 10) \% 3 + (2 \% 3) \right] \% 3$$

$$\left[\begin{array}{l} \downarrow \\ [4 \% 3 \times 10^3 \% 3] \% 3 \\ + \\ [3 \% 3 \times 10^2 \% 3] \% 3 \\ + \\ [7 \% 3 \times 10 \% 3] \% 3 \\ + \\ 2 \% 3 \end{array} \right] \% 3$$

$$\left[(4 \% 3) \% 3 + (3 \% 3) \% 3 + (7 \% 3) \% 3 + (2 \% 3) \right] \% 3$$

$$\Rightarrow [4 \% 3 + 3 \% 3 + 7 \% 3 + 2 \% 3] \% 3$$

↓

$$\underline{\underline{[4 + 3 + 7 + 2] \% 3}}$$

$$[4372] \% 3 \rightarrow \underline{\underline{[4 + 3 + 7 + 2] \% 3}}$$

proved

$$\begin{array}{l} 10^3 \% 3 = 1 \\ 10^2 \% 3 = 1 \\ 10 \% 3 = 1 \end{array}$$

$$\left[\begin{array}{l} (a+b) \% M \\ = (a \% M + b \% M) \% M \end{array} \right]$$

$$\begin{array}{l} (a \times b) \% b \\ = a \% b \end{array}$$

$$\begin{array}{l} (41 \% 10) \% 10 \\ = 1 \% 10 \\ = 1 \\ \text{or, } 41 \% 10 \\ = 1 \end{array}$$

Rule of 4

If No. formed by the last two digit is divisible by 4,
then no. itself is divisible by 4.

$$138 \textcircled{24} \checkmark \rightarrow 13824 \checkmark$$

$$4412 \textcircled{21} \times$$

$$(3484) \% 4 \Rightarrow (84) \% 4$$

↓

$$(3 \times 10^3 + 4 \times 10^2 + 8 \times 10 + 4 \times 10^0) \% 4$$

$$\Rightarrow ((3 \times 10^3) \% 4 + (4 \times 10^2) \% 4 + (8 \times 10) \% 4 + 4 \% 4) \% 4$$

$$\begin{array}{l} \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \\ (3 \% 4 \times \textcircled{10^3 \% 4}) \% 4 \qquad + \qquad (4 \% 4 \times \textcircled{10^2 \% 4}) \% 4 \qquad + \qquad (8 \% 4 \times 10 \% 4) \% 4 \qquad + \qquad 4 \% 4 \\ \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \\ 0 \qquad + \qquad 0 \qquad + \qquad 0 \qquad + \qquad 0 \end{array}$$

$\left[\begin{array}{l} (a \times b) \% M = (a \% M \times b \% M) \% M \\ [(a \times b) \% M + (c \times d) \% M] \% M = ((a \% M \times b \% M) + (c \% M \times d \% M)) \% M \end{array} \right]$

$10^4 \% 4 = 0$
 $10^3 \% 4 = 0$
 $10^2 \% 4 = 0$

$$10 \times 4 \Rightarrow 2$$

$$\Rightarrow \left[\underbrace{(8 \times 4 + 10 \times 4)}_{\text{multiplication}} \times 4 + 4 \times 4 \right] \times 4$$

$$\Rightarrow \left[(8 \times 10) \times 4 + 4 \times 4 \right] \times 4$$

$$\Rightarrow \left[80 \times 4 + 4 \times 4 \right] \times 4$$

$$\Rightarrow \underline{[84]} \times 4 \quad \text{additive rule}$$

$$(3484) \times 4 \rightarrow (84) \times 4$$

Proved

Hw proved $\Rightarrow$ divisibility rule for
5, 8, 9

hook LG

Q2. Given a no. A in the form of an array of size N, and a no. P, return $A \% P$.

ex $\Rightarrow$ $P = 4$ $\Rightarrow$ $A = [1, 2, 3, 4, 4] \rightarrow 12344$

$P = 4$

$A \% P$

$12344 \% 4 = 0$ return

cons.

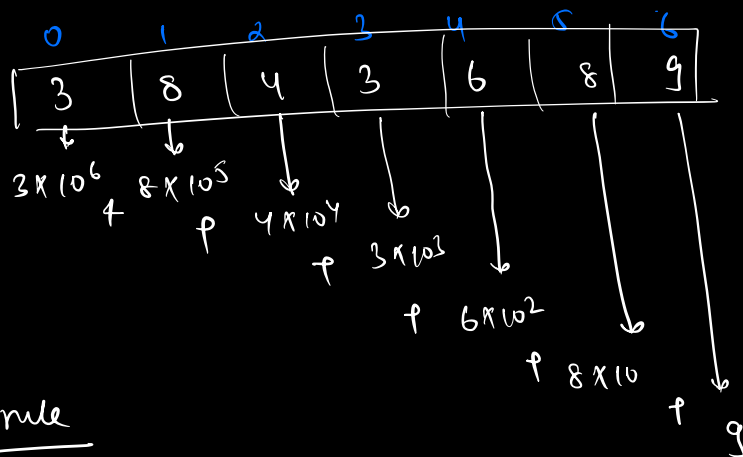
$1 \leq P \leq 10^9$

$1 \leq N \leq 10^5$

long = 10^{18}

$\rightarrow$ 19 digits

ex $\Rightarrow$



i) additive rule

$$\left[(3 \times 10^6) \% P + (8 \times 10^5) \% P + (4 \times 10^4) \% P + (3 \times 10^3) \% P + (6 \times 10^2) \% P + (8 \times 10) \% P + 9 \% P \right] \% P$$

ii) multiplicative rule

$$\left[\begin{aligned} & \underline{(3 \times 10^6 \% p)} \% p + (8 \times 10^5 \% p) \% p + (4 \times 10^4 \% p) \% p \\ & + (3 \times 10^3 \% p) \% p + (6 \times 10^2 \% p) \% p \\ & + (8 \times 10^1 \% p) \% p + 9 \% p \end{aligned} \right] \% p$$

oth digit

$$N \rightarrow (x \times 10^{N-1}) \% p$$

$$N = 10^4$$

$$(10) 10^4 \% p$$

$$\downarrow$$

$$(x \% p \times 10^{N-1} \% p) \% p$$

$$\boxed{[0-9]}$$

easy to
calc

$$\downarrow$$

$$a^x \% p$$

$$\text{power}(a, n, p) \rightarrow \underline{\underline{a^n \% p}}$$

general

$$\begin{array}{cccccccc} [A]_N \Rightarrow & 0 & 1 & 2 & 3 & 4 & \dots & n-2 & n-1 \\ & a_0 & a_1 & a_2 & a_3 & a_4 & \dots & a_{n-2} & a_{n-1} \\ & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow \\ & 10^{N-1} & 10^{N-2} & 10^{N-3} & 10^{N-4} & 10^{N-5} & & 10^1 & 10^0 \end{array}$$

$$\hookrightarrow \left[(a_0 \times 10^{N-1}) + (a_1 \times 10^{N-2}) + (a_2 \times 10^{N-3}) + \dots \right] \% p$$

$\downarrow$

$$\Rightarrow \left[(a_0 \times 10^{N-1}) \% p + (a_1 \times 10^{N-2}) \% p + (a_2 \times 10^{N-3}) \% p + \dots \right] \% p$$

$$\Rightarrow \left(a_0 \% p \times \underbrace{10^{N-1} \% p}_{\text{power}(10, N-1, p)} \right) \% p$$

$$\left(a_1 \% p \times \underbrace{10^{N-2} \% p}_{\text{power}(10, N-2, p)} \right) \% p$$

ans = 0

for (i = 0; i < N; i++) {

ans = ans + (arr[i] % p * power(10, N-1-i, p)) % p

}

return ans

,

$$\left\{ \begin{array}{l} TC \Rightarrow O(N^2) \\ SC \Rightarrow O(1) \end{array} \right.$$

* optimization will discuss in next class

Try it out